GEO Readiness Audit Skill – README

Overview

geo-readiness-audit is a Claude skill that produces comprehensive **Generative Engine Optimization (GEO) readiness reports** for websites and brands. It audits how well a site can be discovered, understood, and recommended by generative AI engines like ChatGPT, Claude, Perplexity, Gemini, and Google AI Overviews. The skill benchmarks the target against competitors and delivers a self-contained, designed HTML report with evidence ledger, competitor analysis, and a 90-day roadmap.

What It Does

The skill automates the creation of **strategic GEO reports** that answer: - Can AI engines find and read your website? - Does AI understand your brand, products, and services? - Will AI recommend you when users ask open-ended questions? - How do you compare to competitors in AI visibility?

It produces reports like the included **SKLUM GEO Readiness Report** example, featuring: - **Hero finding** (headline insight) - **8-dimension competitive benchmark** (or 3–10 dimensions tailored to business type) - **Deterministic technical checks** (robots.txt, schema, canonical, hreflang, etc.) - **Entity knowledge graph** - **90-day improvement roadmap** - **Evidence ledger** (every score traces back to a verifiable observation)

Key Features

✅ **Evidence-first approach**: Every score is computed, traced to an evidence row, or flagged as a judgment
✅ **Competitor benchmarking**: Score your brand against 2–4 rivals
✅ **Measured AI visibility** (optional): Run prompts against real AI engines and compute share-of-voice percentages
✅ **Technical validation**: Automated checks for crawlability, structured data, canonical URLs, hreflang, and more
✅ **Self-contained HTML output**: Beautiful, printable, shareable reports with built-in CSS and inline SVG charts
✅ **Repeatable & updateable**: Keep ledger IDs stable to refresh reports over time

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Dependencies

Python 3.7+ (Required)

The skill includes **three Python scripts** that handle deterministic tasks:

1. **`scripts/technical_checks.py`**
   - Crawls robots.txt, sitemaps, and named pages (raw HTML only, no JavaScript)
   - Validates schema markup, canonical URLs, hreflang, AI crawler access
   - Outputs JSON that merges into the report
2. **`scripts/aggregate_visibility.py`**
   - Converts a CSV log of AI engine answers into visibility percentages
   - Computes share-of-voice metrics
3. **`scripts/render_report.py`**
   - Validates the report JSON
   - Renders the final self-contained HTML with radar charts, entity trees, and tables
   - Supports logo embedding and strict validation mode

Important: These scripts use **standard library only** (no pip installs needed).

Installation & Setup

1. **Extract the skill file**
   Unzip `geo-readiness-audit.skill` to a working directory.

2. **Verify Python 3**

   ```
   python3 --version
   ```

   Ensure you have Python 3.7 or newer.

3. **Test the scripts**

   ```
   python3 scripts/render_report.py assets/example_sklum.json --check
   ```

   You should see: `OK: X warning(s).`

Usage Workflow

1. Trigger the Skill in Claude

Use natural language to request a GEO audit: - *“Do a GEO audit for example.com”* - *“Audit how well [brand] will be recommended by AI search engines”* - *“Benchmark [domain] against [competitor1, competitor2] for AI visibility”* - *“Refresh the GEO report for [site] with updated technical checks”*

2. Claude Collects Evidence

The skill guides Claude to: - Identify the domain, brand, and 2–4 competitors - Run **`technical_checks.py`** to verify crawlability, schema, canonical, etc. - Collect evidence from pages, search results, and third-party mentions - Score 3–10 dimensions (e.g., Answerability, Brand Authority, Sustainability) - Build an evidence ledger with URLs, confidence levels, and methods

3. (Optional) Measure AI Visibility

If you want **measured percentages** (not estimates): - Claude provides a prompt pack (20–30 questions in the brand’s market) - **You run the prompts** against ChatGPT, Perplexity, Gemini, etc., and log answers to CSV - Run **`aggregate_visibility.py`** to compute mentioned/recommended/linked percentages - Merge the output back into the report

4. Render the HTML Report

Claude writes a structured JSON file, then calls:

```
python3 scripts/render_report.py report.json \
  --merge-technical technical.json \
  [--merge-visibility visibility.json] \
  [--logo logo.svg] \
  -o report.html
```

The final **`report.html`** is a self-contained, print-ready document with: - Hero scores and headline finding - Radar chart and competitive table - Entity map (SVG tree) - Technical checklist with pass/warn/fail status - Evidence ledger (every claim linked to a source) - 90-day roadmap with specific actions

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File Structure

```
geo-readiness-audit/
├── SKILL.md                        # Skill metadata and instructions for
Claude
├── scripts/
│   ├── technical_checks.py         # Crawl & validate technical GEO (Python
3)
```

```
│   ├── aggregate_visibility.py     # Compute AI visibility percentages
(Python 3)
│   └── render_report.py            # Validate & render HTML report (Python
3)
├── references/
│   ├── rubric.md                   # Scoring anchors, dimension presets,
evidence rules
│   ├── report-schema.md            # JSON schema reference for all report
fields
│   └── ai-visibility-prompts.md    # How to run & report measured visibility
studies
├── assets/
│   ├── example_sklum.json          # Full worked example (consumer e-
commerce)
│   ├── example_minimal_b2b.json    # Minimal B2B example
│   └── report.css                  # Design system (embedded by render
script)
```

Example Commands

Run technical checks

```
python3 scripts/technical_checks.py https://www.example.com \
  --brand "Example" \
  --page home=https://www.example.com/ \
  --page product=https://www.example.com/product/123 \
  --sample-sitemap 6 \
  -o technical.json
```

Aggregate visibility study

```
python3 scripts/aggregate_visibility.py results.csv \
  --engines "ChatGPT, Perplexity" \
  --date 2026-10-12 \
  --market "Spain, Spanish, logged out" \
  -o visibility.json
```

Render the report

```
python3 scripts/render_report.py report.json \
  --merge-technical technical.json \
  --merge-visibility visibility.json \
  --logo logo.svg \
  -o report.html
```

Validate only (no HTML output)

```
python3 scripts/render_report.py report.json --check
```

Key Principles (Built Into the Skill)

1. **A number is computed, traced, or flagged**
   No unsupported claims. Every score points to evidence rows or is marked as a judgment.

2. **Unverified ≠ Failed**
   If technical checks are blocked (network, proxy, bot protection), they're marked `unverified`, not `fail`.

3. **Honest asymmetry**
   If competitors were sampled less deeply, the report says so in `benchmark.note`.

4. **No placeholder guesses**
   If AI visibility wasn't measured, the section is left out—never filled with estimates.

5. **Repeatable & updateable**
   Keep ledger IDs stable across refreshes so earlier citations still resolve.

Troubleshooting

Scripts fail with "network refused" or HTTP 403

- This is expected in sandboxed environments (proxies, firewalls, corporate networks).
- The script marks checks as `unverified` and tells you to re-run from a machine with open access.
- Merge the `technical.json` later with `--merge-technical`.

Warnings about missing evidence or gaps

- The validator warns when dimension scores differ by 10+ points without cited evidence.
- Fix by adding `evidence: ["E1", "E2"]` in `benchmark.basis.<dimension>` or mark `derived: true`.

Logo not showing

- Use `--logo logo.svg` (or `.png`, `.jpg`, `.webp`, max 300 KB).
- The script embeds it as a base64 data URI.

When to Use This Skill

✅ Client asks: *"Will AI recommend us?"*
✅ RFP requires competitive AI visibility analysis

✅ You need a repeatable, evidence-backed SEO/GEO audit
✅ Benchmarking a brand against 2–4 competitors for generative search
✅ Periodic monitoring (run monthly/quarterly with the same prompts)

Limitations

⚠️ **Requires Python 3** for technical checks, aggregation, and rendering
⚠️ **Not a real-time crawler**: It's a sample-based audit, not a full site crawl
⚠️ **No JavaScript rendering**: Checks raw HTML only (like many AI crawlers)
⚠️ **Measured visibility is manual**: You must run prompts yourself and log answers to CSV
⚠️ **Network-dependent**: Corporate proxies/firewalls may block `technical_checks.py`

Output Example

See `assets/example_sklum.json` (consumer e-commerce, 8 dimensions) or `assets/example_minimal_b2b.json` (B2B SaaS, 5 dimensions) for complete worked examples. Render them with:

```
python3 scripts/render_report.py assets/example_sklum.json -o
sklum_report.html
```

Support & Feedback

- **Skill version**: 1.0
- **Maintained by**: Benoit Bousquié
- **Questions?**: Benoit Bousquié
